Dhivyuga - Deep Feature & CRUD Analysis

Complete inventory of all features, CRUD operations, API endpoints, database tables, and backend (Supabase) touchpoints before Appwrite migration.

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1. Database Schema: 11 Tables

1.1 categories

Field	Type	Notes
id	UUID (PK)	Auto-generated
name	VARCHAR(255)	NOT NULL
description	TEXT	Nullable
is_active	BOOLEAN	Default true
created_at	TIMESTAMPTZ	Default NOW()

Relationships: has_many mantras

Seed Data: 10 records - Wealth, Protection, Health, Wisdom, Love, Success, Peace, Spiritual Growth, Obstacle Removal, Good Fortune

1.2 deities

Field	Type	Notes
id	UUID (PK)	Auto-generated
name	VARCHAR(255)	NOT NULL
description	TEXT	Nullable
sanskrit_name	VARCHAR	For Grahas
day_of_week	VARCHAR	For Grahas
color	VARCHAR	For Grahas
gemstone	VARCHAR	For Grahas
metal	VARCHAR	For Grahas
element	VARCHAR	For Grahas
direction	VARCHAR	For Grahas
is_active	BOOLEAN	Default true
created_at	TIMESTAMPTZ	Default NOW()

Relationships: has_many mantras

Seed Data: 10 deities (Ganesha, Lakshmi, Saraswati, Hanuman, Shiva, Vishnu, Devi, Krishna, Rama, Kali) + 9 Grahas (Surya, Chandra, Mangal, Budh, Guru, Shukra, Shani, Rahu, Ketu)

1.3 recitation_counts

Field	Type	Notes
id	UUID (PK)	Auto-generated
count_value	INTEGER	NOT NULL
description	TEXT	Nullable
created_at	TIMESTAMPTZ	Default NOW()

Relationships: has_many mantras

Seed Data: 6 records - 11, 27, 54, 108, 1008, 10008

1.4 recitation_times

Field	Type	Notes
id	UUID (PK)	Auto-generated
name	VARCHAR(255)	NOT NULL
description	TEXT	Nullable
created_at	TIMESTAMPTZ	Default NOW()

Relationships: has_many mantras

Seed Data: 4 records - Morning, Noon, Evening, Brahma Muhurta

1.5 kalams

Field	Type	Notes
id	UUID (PK)	Auto-generated
name	VARCHAR(255)	NOT NULL
planet	VARCHAR(100)	Nullable
description	TEXT	Nullable
is_auspicious	BOOLEAN	Default true
created_at	TIMESTAMPTZ	Default NOW()

Relationships: has_many mantras

Seed Data: 7 records - Rahu Kalam, Yama Kalam, Gulika Kalam (inauspicious); Surya Hora, Chandra Hora, Guru Hora, Shukra Hora (auspicious)

1.6 time_ranges

Field	Type	Notes
id	UUID (PK)	Auto-generated
start_time	TIME	NOT NULL
end_time	TIME	NOT NULL
description	TEXT	Nullable
created_at	TIMESTAMPTZ	Default NOW()

Relationships: has_many mantras

Seed Data: 4 records - 3:30-5:30 AM (Brahma Muhurta), 6:00-9:00 AM (Morning), 11:30-12:30 PM (Noon), 6:00-8:00 PM (Evening)

1.7 mantras (Core Entity)

Field	Type	Notes
id	UUID (PK)	Auto-generated
title	VARCHAR(500)	NOT NULL
text	TEXT	NOT NULL
category_id	UUID (FK)	References categories
deity_id	UUID (FK)	References deities
count_id	UUID (FK)	References recitation_counts
time_id	UUID (FK)	References recitation_times
kalam_id	UUID (FK)	References kalams
range_id	UUID (FK)	References time_ranges
view_count	INTEGER	Default 0
search_vector	tsvector	Full-text search index
created_at	TIMESTAMPTZ	Default NOW()

Relationships: belongs_to category, deity, count, time, kalam, range; has_many descriptions, translations

Seed Data: 2+ sample mantras (Om Gam Ganapataye Namaha, Om Shreem Mahalakshmiyei Namaha)

1.8 mantra_descriptions

Field	Type	Notes
id	UUID (PK)	Auto-generated
mantra_id	UUID (FK)	References mantras, ON DELETE CASCADE
meaning	TEXT	NOT NULL
transliteration	TEXT	Nullable
pronunciation_guide	TEXT	Nullable
benefits	TEXT[]	Array of benefits
historical_context	TEXT	Nullable
spiritual_significance	TEXT	Nullable
usage_instructions	TEXT	Nullable
precautions	TEXT	Nullable
related_scriptures	TEXT[]	Array of scripture references
created_at	TIMESTAMPTZ	Default NOW()
updated_at	TIMESTAMPTZ	Auto-updated via trigger

Relationships: belongs_to mantra

1.9 mantra_translations

Field	Type	Notes
id	UUID (PK)	Auto-generated
mantra_id	UUID (FK)	References mantras
language_id	UUID (FK)	References languages
text	TEXT	NOT NULL
transliteration	TEXT	Nullable
pronunciation_guide	TEXT	Nullable
meaning	TEXT	Nullable
benefits	TEXT[]	Array
usage_notes	TEXT	Nullable
created_at	TIMESTAMPTZ	Default NOW()
updated_at	TIMESTAMPTZ	Default NOW()

Relationships: belongs_to mantra, language

1.10 languages

Field	Type	Notes
id	UUID (PK)	Auto-generated
code	VARCHAR	e.g., en, hi, ta, sa
name	VARCHAR	e.g., English, Hindi
native_name	VARCHAR	e.g., Tamil, Hindi
direction	VARCHAR	ltr or rtl
sort_order	INTEGER	Display ordering
is_active	BOOLEAN	Default true
created_at	TIMESTAMPTZ	Default NOW()
updated_at	TIMESTAMPTZ	Default NOW()

Relationships: has_many translations

Seed Data: English, Hindi, Tamil, Sanskrit

1.11 auth.users (Supabase Auth)

Field	Type	Notes
id	UUID (PK)	Supabase managed
email	VARCHAR	Admin email
password	HASHED	Supabase managed

Field	Type	Notes
user_metadata	JSONB	{ role: 'admin', name: 'Dhivyuga Admin' }

Seed Data: 1 admin user (admin@dhivyuga.com)

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2. PostgreSQL-Specific Features

- **tsvector** column on mantras for full-text search
- **GIN index** on search_vector for fast text search
- **3 Database Triggers:**

1. `mantras_search_vector_update` - Auto-updates search_vector on mantra insert/update
2. `mantra_descriptions_updated_at` - Auto-updates updated_at timestamp
3. `mantra_descriptions_search_update` - Propagates description changes to mantra search_vector

- **5 Indexes:**

1. `mantras_search_idx` (GIN on search_vector)
2. `idx_mantras_view_count` (DESC on view_count)
3. `idx_mantras_created_at` (DESC on created_at)
4. `idx_mantra_descriptions_mantra_id` (on mantra_id FK)
5. `idx_mantra_descriptions_updated_at` (DESC on updated_at)

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3. API Routes: 17 Endpoints

3.1 Search & Discovery

#	Route	Method	Purpose	Tables
1	`/api/search`	GET	Full-text search + multi-filter	mantras + 6 joins
2	`/api/autocomplete`	GET	Prefix-match suggestions (ILIKE)	mantras, deities, categories
3	`/api/filters`	GET	All filter dropdown options	categories, deities, recitation_times, kalams

3.2 Mantra Endpoints

#	Route	Method	Purpose	Tables
4	`/api/mantras/trending`	GET	Top 6 by view_count DESC	mantras + deities, categories
5	`/api/mantras/[id]`	GET	Full detail + translations + related	mantras + 7 joins + translations + related
6	`/api/mantras/[id]/view`	POST	Increment view counter	mantras (select + update)

#	Route	Method	Purpose	Tables
7	`/api/mantras/[id]/translations`	GET	List translations	mantra_translations + languages
8	`/api/mantras/[id]/translations`	POST	Create translation	mantra_translations
9	`/api/mantras/[id]/translations`	PUT	Update translation	mantra_translations
10	`/api/mantras/[id]/translations`	DELETE	Delete translation	mantra_translations

3.3 Reference Data

#	Route	Method	Purpose	Tables
11	`/api/categories`	GET	List active categories	categories
12	`/api/categories`	POST	Create category	categories
13	`/api/deities`	GET	List active deities	deities
14	`/api/languages`	GET	List active languages	languages
15	`/api/languages`	POST	Create language	languages

3.4 Admin & Seeding

#	Route	Method	Purpose	Tables
16	`/api/seed-grahas`	POST	Seed 9 planetary deities (idempotent)	deities
17	`/api/create-admin`	POST	Create admin user account	auth.users

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4. CRUD Operations by Entity

4.1 Mantras

Operation	Location	Method
Create	Admin `/admin/mantras`	Direct Supabase `insert()` client-side
Read (list)	Admin `/admin/mantras`	Direct Supabase `select()` with joins client-side
Read (detail)	Public `/mantra/[id]`	API route `/api/mantras/[id]`
Read (trending)	Home page	API route `/api/mantras/trending`
Read (search)	Search page	API route `/api/search`
Update	Admin `/admin/mantras`	Direct Supabase `update()` client-side
Delete	Admin `/admin/mantras`	Direct Supabase `delete()` client-side
View count++	Public detail page	API route `/api/mantras/[id]/view`

4.2 Categories

Operation	Location	Method
Create	Admin `/admin/categories`	Direct Supabase `insert()` client-side
Read	Admin + Public (home, filters)	Client-side + API `/api/categories`
Update	Admin `/admin/categories`	Direct Supabase `update()` client-side
Delete	Admin `/admin/categories`	Direct Supabase `delete()` client-side

4.3 Deities

Operation	Location	Method
Create	Admin `/admin/deities`	Direct Supabase `insert()` client-side
Read	Admin + Public (home, filters, grahas)	Client-side + API `/api/deities`
Update	Admin (incl. toggle active)	Direct Supabase `update()` client-side
Delete	Admin `/admin/deities`	Direct Supabase `delete()` client-side

4.4 Mantra Translations

Operation	Location	Method
Create	Admin LanguageManager	API POST `/api/mantras/[id]/translations`
Read	Admin + Public detail	API GET `/api/mantras/[id]/translations`
Update	Admin LanguageManager	API PUT `/api/mantras/[id]/translations`
Delete	Admin LanguageManager	API DELETE `/api/mantras/[id]/translations`

4.5 Settings (4 sub-entities from `/admin/settings`)

Entity	C	R	U	D	Method
Recitation Times	Yes	Yes	Yes	Yes	Direct Supabase client-side
Recitation Counts	Yes	Yes	Yes	Yes	Direct Supabase client-side
Kalams	Yes	Yes	Yes	Yes	Direct Supabase client-side
Time Ranges	Yes	Yes	Yes	Yes	Direct Supabase client-side

4.6 Languages

Operation	Location	Method
Create	API only	POST `/api/languages`

Operation	Location	Method
Read	Admin LanguageManager	GET `/api/languages`

4.7 Authentication

Operation	Location	Method
Create admin	Setup page `/admin/setup`	API POST `/api/create-admin` (Supabase Admin Auth)
Login	`/admin/login`	`supabase.auth.signInWithPassword()`
Logout	Admin sidebar	`supabase.auth.signOut()`
Session check	AuthGuard component	`supabase.auth.getSession()` + `onAuthStateChange`

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5. Public-Facing Features

#	Feature	Page/Component	Data Source	Browser API
1	Full-text search	`/search` + SearchBar	`/api/search` (tsvector)	-
2	Autocomplete suggestions	SearchBar	`/api/autocomplete` (ILIKE prefix)	-
3	Quick filter chips	QuickFilters	Hardcoded (Wealth, Protection, Health, Wisdom)	-
4	Advanced filtering	SearchFilters sidebar	`/api/filters`	-
5	Trending mantras	TrendingMantras	`/api/mantras/trending`	-
6	Browse by category	CategoryGrid	`/api/categories`	-
7	Browse by deity	DeityGrid	`/api/deities`	-
8	Mantra detail view	`/mantra/[id]`	`/api/mantras/[id]`	-
9	Multi-language translations	MantraDisplay	Embedded in mantra detail	-
10	Text-to-Speech	MantraCard + MantraDisplay	-	Web Speech API
11	Share mantra	MantraCard + MantraDisplay	-	Navigator.share API
12	Copy to clipboard	MantraDisplay	-	Clipboard API
13	View counter	Mantra detail page	`/api/mantras/[id]/view`	-
14	Navagraha planetary info	`/grahas`	Direct Supabase query	-
15	Contact form	`/contact`	Simulated (no backend)	-
16	About page	`/about`	Static content	-
17	Terms page	`/terms`	Static content	-

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6. Admin Features

#	Feature	Page	Key Components
1	Dashboard with stats	`/admin`	DashboardCards, DataTable
2	Mantra CRUD + translations	`/admin/mantras`	Dialog, Tabs, LanguageManager, WysiwygEditor
3	Category CRUD	`/admin/categories`	Card grid, Dialog, Framer Motion
4	Deity CRUD + toggle active	`/admin/deities`	Card grid, Dialog, Framer Motion
5	Settings CRUD (4 tables)	`/admin/settings`	4 Tabs, Tables, Dialog
6	Analytics dashboard	`/admin/analytics`	Metric cards, top mantras, progress bars
7	Admin login	`/admin/login`	AuthContext, password toggle
8	Admin setup	`/admin/setup`	Creates initial admin user
9	Sidebar navigation	All admin pages	AdminLayout, responsive menu
10	Auth protection	All admin pages	AuthGuard component

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7. Supabase Touchpoints: 26 Files

#	File	Usage
1	`lib/supabase.ts`	Client init + TypeScript types
2	`contexts/auth-context.tsx`	signInWithPassword, signOut, getSession, onAuthStateChange
3	`app/api/search/route.ts`	textSearch, select with joins, eq filters
4	`app/api/autocomplete/route.ts`	ilike prefix queries on 3 tables
5	`app/api/filters/route.ts`	select from 4 tables
6	`app/api/mantras/trending/route.ts`	select with joins, order by view_count
7	`app/api/mantras/[id]/route.ts`	select with 7 joins + translations + related
8	`app/api/mantras/[id]/view/route.ts`	select + update (increment counter)
9	`app/api/mantras/[id]/translations/route.ts`	Full CRUD on mantra_translations
10	`app/api/categories/route.ts`	select + insert
11	`app/api/deities/route.ts`	select with is_active filter
12	`app/api/languages/route.ts`	select + insert
13	`app/api/seed-grahas/route.ts`	ilike check + insert 9 records
14	`app/api/create-admin/route.ts`	auth.admin.createUser()
15	`app/admin/page.tsx`	select with counts (dashboard stats)
16	`app/admin/mantras/page.tsx`	Full CRUD with joins (client-side)
17	`app/admin/categories/page.tsx`	Full CRUD with mantra count (client-side)

#	File	Usage
18	`app/admin/deities/page.tsx`	Full CRUD + toggle active (client-side)
19	`app/admin/settings/page.tsx`	CRUD on 4 tables (client-side)
20	`app/admin/analytics/page.tsx`	select for stats
21	`app/admin/login/page.tsx`	Uses AuthContext (indirect)
22	`app/admin/setup/page.tsx`	Calls /api/create-admin
23	`app/grahas/page.tsx`	Direct select + ilike filters
24	`components/admin/language-manager.tsx`	CRUD on translations + languages
25	`supabase/schema.sql`	Full DDL + triggers + seed data
26	`package.json`	@supabase/supabase-js dependency

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8. Tech Stack Summary

Layer	Technology
Frontend Framework	Next.js 15.5.2 (App Router)
UI Library	React 18.2.0
Styling	TailwindCSS 3.3.0 + shadcn/ui + Radix UI
Animations	Framer Motion 12.23.12
Icons	Lucide React 0.542.0
Forms	React Hook Form 7.47.0 + Zod
Rich Text Editor	React Quill New 3.3.0
Database	Supabase (PostgreSQL)
Authentication	Supabase Auth (email/password)
Search	PostgreSQL Full-Text Search (tsvector + GIN)
Deployment	Netlify (with @netlify/plugin-nextjs)
Language	TypeScript 5

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